Capstone Project-23

Capstone Project Case Study on Pollution in India with five interconnected sections (Advanced Excel, SQL, Python, Data Science, Power BI). The theme is **analyzing air and water pollution trends in India and proposing solutions**. Each section builds on the previous one, and the final deliverable is a **Power BI dashboard** that integrates all outputs.

Real-Time Case Study Theme

Pollution in India: Air Quality, Water Contamination, and Policy Impact (2010–2025)
Dataset: CPCB (Central Pollution Control Board) air quality index (AQI), water contamination levels (rivers, groundwater), industrial emissions, vehicular data, population density, and government policy interventions.

Section 1: Advanced Excel

Focus: Initial exploration and cleaning of pollution data.
10 Questions/Tasks

1. Import AQI dataset into Excel. Use PivotTables to summarize AQI by city.
2. Apply conditional formatting to highlight cities with AQI > 200 (poor quality).
3. Create a trendline chart showing AQI changes from 2010–2025.
4. Use VLOOKUP to merge industrial emission data with AQI records.
5. Build a calculated column for "Pollution per capita = Emission / Population."
6. Apply data validation to restrict input errors in water contamination levels.
7. Use advanced filters to isolate top 10 polluted rivers.
8. Create a dynamic chart with slicers for year and pollutant type.
9. Perform scenario analysis: What if vehicular emissions reduce by 20%?
10. Export the cleaned dataset for SQL integration.

Section 2: SQL

Focus: Structured queries for deeper insights.
10 Questions/Tasks

1. Write a query to calculate average AQI per state.
2. Create a view Critical Zones for cities with AQI > 300 and water contamination > permissible limits.
3. Query top 5 industries contributing to emissions.
4. Use window functions to rank states by AQI improvement rate.

5. Join AQI and population datasets to analyse correlation.
6. Create a stored procedure to filter pollution data by year range.
7. Query average water contamination levels in rivers across states.
8. Build a temporary table for "Pollution vs GDP" comparison.
9. Use GROUP BY to calculate average AQI per season (summer, monsoon, winter).
10. Export SQL results into CSV for Python analysis.

Section 3: Python

Focus: Data wrangling, visualization, and scripting.
10 Questions/Tasks

1. Import SQL output into Pandas Data Frame.
2. Clean missing values in AQI and water datasets.
3. Generate correlation matrix between AQI, industrial emissions, and population density.
4. Plot line graph of AQI trends using Matplotlib.
5. Create a bar chart comparing water contamination by river.
6. Write a script to calculate year-over-year AQI changes.
7. Use Pandas group by to summarize pollution by industry sector.
8. Export processed dataset into Excel for Data Science section.
9. Automate pollution report generation with Python script.
10. Save outputs in CSV for Power BI integration.

Section 4: Data Science (Non-ML Focus)

Focus: Statistical analysis and hypothesis testing.
10 Questions/Tasks

1. Perform correlation analysis between industrial emissions and AQI.
2. Conduct hypothesis test: "Cities with >50% public transport usage have lower AQI."
3. Use regression (without ML) to estimate linear relationship between vehicular density and AQI.
4. Calculate moving averages for AQI trends.
5. Perform chi-square test on categorical data (policy adoption vs AQI improvement).
6. Build time-series decomposition for AQI data.
7. Conduct variance analysis (ANOVA) for AQI across states.

8. Estimate confidence intervals for average water contamination levels.
9. Perform sensitivity analysis on emission reduction scenarios.
10. Export statistical summaries for Power BI.

Section 5: Power BI

Focus: Interactive dashboard integrating all outputs.
10 Questions/Tasks

1. Import datasets from Excel, SQL, and Python outputs.
2. Build a heatmap showing AQI hotspots by city.
3. Create a line chart of AQI trends (2010–2025).
4. Add slicers for year, pollutant type, and industry sector.
5. Create KPI cards: "Avg AQI," "Water Contamination Index."
6. Build a scatter plot: AQI vs Population Density.
7. Add drill-through to industry-level emission details.
8. Integrate statistical outputs (confidence intervals, correlations).
9. Create scenario simulation: "If vehicular emissions reduce by 20%."
10. Publish final dashboard for stakeholder review.

Flow of Interconnection

- **Excel** → Data cleaning and exploration.
- **SQL** → Structured queries refine datasets.
- **Python** → Visualization and automation.
- **Data Science** → Statistical validation.
- **Power BI** → Final interactive dashboard with pollution insights and solutions.

This gives you a **real-time case study on pollution in India**, with **10 practical questions per section**, all interconnected, ending in a **Power BI dashboard** that highlights pollution hotspots, causes, and potential interventions.